

Why This Study Evaluates Chronologically

The argument, the published support, and our own controlled measurement

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1 The claim

This study evaluates on a *chronological* split — train on the earliest transactions, test on the latest — rather than the random stratified split used by most published work on the ULB dataset.

The justification is a **validity** argument, not a performance one. A chronological split is correct because it is the only protocol whose question has a deployment counterpart. Whether it also produces lower scores is a separate, empirical matter, addressed in Section 4.

2 Why a random split asks the wrong question

A random split assigns transactions to train and test by coin flip. The model is therefore fitted on transactions that occurred *after* the ones it is scored on. No deployed fraud system can do this: at scoring time, the future has not happened yet. Three consequences are measurable directly on ULB.

(a) The distribution moves, even inside 48 hours. ULB spans exactly 48 hours. Across our chronological partitions the fraud prior falls by 37% relative and the mean transaction amount by 19%:

Partition	Window (h)	Frauds	Fraud rate	Mean amount
Train	0.00–36.92	384	0.193%	89.77
Validation	36.92–42.04	56	0.131%	97.52
Test	42.04–48.00	52	0.122%	72.54

The window also covers two full day–night cycles with a $35.5\times$ ratio between the peak and trough hourly fraud rates (02:00 peaks at 1.71%). Shuffling destroys all of this structure by construction: it makes the train and test distributions identical by definition, which is precisely the condition a deployed model does not enjoy.

(b) Shuffling separates duplicate transactions into train and test. ULB contains 1,081 exact duplicate rows. All 773 duplicate groups share a single Time value, so chronological sorting places every copy adjacent to its twin and they always land in the same partition. Measured across all three boundaries:

	Train→Test	Train→Val	Val→Test	Total leaked
Chronological split	0	0	0	0
Random split	249	207	61	517

Under shuffling the model is fitted on a row and then scored on a byte-identical copy of it. That is memorisation being reported as generalisation, and the chronological protocol is immune to it for free — without deleting any data.

(c) **The resulting estimate has no operational meaning.** A random-split score answers “how well would this model rank transactions if it had already seen the future?” That number cannot be used to size a review queue, set an alert threshold, or forecast caseload, because the condition it assumes never holds in production.

3 What the published literature says

The four sources below support the *principle*. We deliberately do not compare our absolute scores to theirs — see Section 5.

- **Dal Pozzolo, Boracchi, Caelen, Alippi and Bontempi (2018)**, *Credit Card Fraud Detection: A Realistic Modeling and a Novel Learning Strategy*, IEEE Transactions on Neural Networks and Learning Systems 29(8):3784–3797.
<https://re.public.polimi.it/bitstream/11311/1044896/1/08038008.pdf>
These are the authors of the ULB dataset itself. They formalise fraud detection around three operating conditions — *concept drift* (customer habits evolve and fraudsters change strategy over time), class imbalance, and *verification latency* (only a small set of transactions is checked in time) — and argue that a realistic evaluation must reflect them. Concept drift and verification latency are both inherently temporal; a shuffled split represents neither.
- **Sang (2026)**, *Enhancing Credit Card Fraud Detection under Severe Class Imbalance using Cost-Sensitive Learning and Threshold Optimization*, EAI Endorsed Transactions on ISMLA, Vol. 3.
<https://publications.eai.eu/index.php/ismla/article/view/12078>
The only study we located that evaluates ULB under *both* protocols. It reports that the chronological split yields lower PR-AUC than the random one and concludes that “random stratification may provide optimistic estimates for this benchmark dataset when earlier and later transactions are mixed across training, validation, and test partitions.” It also lists, among the gaps motivating the work, that “random stratified validation can mix earlier and later transactions, potentially overstating future-period performance,” and describes its own chronological experiment as evaluating “deployment realism.”
- **Data Leakage and Deceptive Performance: A Critical Examination of Credit Card Fraud Detection Methodologies** (2025), arXiv:2506.02703.
<https://arxiv.org/abs/2506.02703>
An audit of published ULB pipelines. It identifies four recurring flaws, of which the third is “*inadequate temporal validation for transaction data*.” It demonstrates the stakes by constructing a deliberately leaky minimal MLP that reaches 99.9% recall and outperforms far more sophisticated published methods, concluding that “proper evaluation methodology matters more than model complexity in fraud detection research.”
- **Khaleghpour and McKinney**, *Leakage Safe Graph Features for Interpretable Fraud Detection in Temporal Transaction Networks*, arXiv:2603.06632.
<https://arxiv.org/abs/2603.06632>
Applies a strict chronological split to the Elliptic dataset — a different dataset, cited here only for the principle — on the grounds that it “reflects a realistic deployment setting in which models are trained on historical data and evaluated on future transactions.” It observes that the illicit rate decreases across time, “indicating temporal distribution shift that can affect generalization,” and reports a substantial drop from validation to the held-out future test period.

4 Our own controlled measurement

The literature establishes the principle. To measure the *size* of the effect we compare our own models against themselves, holding everything fixed — same data, same features, same code, same hyperparameters, same threshold rule — and changing only how rows are assigned to partitions. This is the only comparison in which the split is the sole variable.

Table 1: Chronological versus random, everything else held constant.

Configuration	chronological	random	random – chrono
<i>at 70/15/15</i>			
XGBoost + class weight	0.7688	0.8372	+0.0684
XGBoost + SMOTE-ENN	0.7766	0.8104	+0.0338
CatBoost + GAN	0.7804	0.8399	+0.0595
MLP raw	0.7794	0.8267	+0.0473
LSTM + class weight	0.7803	0.7965	+0.0162
RF + SMOTE-ENN	0.7947	0.8120	+0.0173
GraphSAGE + weight	0.7389	0.6948	−0.0441
<i>mean +0.0283, positive in 6 of 7</i>			
<i>at 66/14/20</i>			
XGBoost + class weight	0.8055	0.8291	+0.0236
XGBoost + SMOTE-ENN	0.8102	0.8020	−0.0082
RF + SMOTE-ENN	0.8224	0.7907	−0.0317
<i>mean −0.0054, positive in 1 of 3</i>			

Pooling all ten paired measurements:

$$\text{random} - \text{chronological} = +0.0182, \quad 95\% \text{ CI } [-0.0084, +0.0448], \quad \text{positive in } 7/10.$$

Honest reading. The direction agrees with theory and with every source in Section 3, and the point estimate is positive. But the interval includes zero ($t = 1.55$, one-sided sign test $p = 0.172$), and the estimate is not stable across cut points. **Our data show the direction, not a reliable magnitude.** Two further notes bound the interpretation:

- Our random arm carries the 517 leaked duplicate rows documented above, which inflates it. The measurement is therefore biased *towards* finding a larger effect, not a smaller one.
- Our random arm applies all resampling *inside* the fitted pipeline, so it captures the ordering effect only. It excludes the resample-before-split channel that arXiv:2506.02703 identifies as common in published work, which is the larger of the two.

Neither of these changes the conclusion of Section 1: the case for chronological evaluation rests on validity, and would stand even if the two protocols scored identically.

5 Why we do not match numbers with the published studies

Absolute PR-AUC figures are not comparable across these papers, and this study does not treat them as though they were. The reasons are concrete:

- **Different architectures and objectives.** Sang (2026) uses cost-sensitive XGBoost and an MLP trained with weighted BCE plus Focal Loss, with synthetic oversampling deliberately excluded. This study’s champion is a Random Forest with SMOTE-ENN. These are different models solving the problem in different ways.

- **Different search budgets and spaces.** Sang tunes with Optuna over tree depth, minimum child weight, learning rate, subsampling ratios, regularisation coefficients and the positive-class weight; the trial count is not stated. This study uses 30 Optuna trials by default, and on the random arms varied only that one weighting parameter over a six-point grid. A difference in score between the two is at least as likely to reflect search effort as model quality.
- **Undocumented preprocessing.** Several studies do not state where resampling sits relative to the split, or whether duplicates were removed. Those choices move the metric more than the model does, and cannot be reconstructed from the papers.
- **Different data.** Some headline figures are obtained on private datasets with far more fraud cases, where interval widths are an order of magnitude tighter for reasons of sample size alone.
- **Measurement precision.** On this test set a single transaction is worth ± 0.020 PR-AUC and bootstrap intervals are 0.196–0.236 wide. Differences of the size usually discussed between papers are smaller than the error bars on either side of the comparison.

For these reasons the evidence offered here is Table 1, in which the split is the only thing that changes, rather than any comparison of this study’s absolute scores against a published leaderboard.

6 Conclusion

1. Chronological evaluation is used because it is the only protocol whose question a deployed system could answer. This holds regardless of the scores it produces.
2. On ULB specifically, shuffling destroys measurable temporal structure and separates 517 duplicate rows across the train/test boundary; the chronological split leaks none.
3. The dataset’s own authors, and three subsequent studies, all argue for temporally faithful evaluation on the same grounds.
4. Our controlled measurement puts the effect at $+0.018$ PR-AUC with a confidence interval that includes zero. The direction is consistent; the magnitude is not established at this sample size, and we do not claim it is.